

Problems in combinatorics

Counting without listing — the two rules, and the three formulas that come from them.

LESSON 60–62

3 × 40 MIN

ALGEBRA 11, §4.1

EXTENSION (P&S 1)

LESSON CLOCK

18'	26'	30'	28'	30'	8'
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The two rules	18 min
Permutations	26 min
Combinations	30 min
Does order matter?	28 min
Practice	30 min
Homework	8 min

LEARNING OBJECTIVES

- Apply the addition and multiplication rules of counting.
- Compute permutations, with and without repetition.
- Compute combinations and use the symmetry property.
- Decide whether order matters in a given problem.

TEXTBOOK

National	pp. 273–290
Cambridge	Probability & Statistics 1, pp. 82–96
Workbook	Combinatorics sheet C1

01 Explanation

The two rules

MULTIPLICATION (“AND”)

If one choice can be made in m ways and, independently, another in n ways, the pair can be made in $m \cdot n$ ways.

ADDITION (“OR”)

If a choice can be made in m ways **or**, in a mutually exclusive way, in n ways, there are $m + n$ ways in total.

Every formula below is these two rules applied repeatedly. Reading the word “and” or “or” in the problem is most of the work.

Example. A menu has 4 starters, 6 mains and 3 desserts. A full meal: $4 \times 6 \times 3 = 72$.
Just one course: $4 + 6 + 3 = 13$.

Permutations

$$n! = n(n-1)(n-2)\dots 2 \cdot 1 \quad 0! = 1$$

QUESTION	FORMULA	EXAMPLE
arrange all n in a row	$n!$	5 books: 120
arrange k out of n , order matters	$A(n,k) = \frac{n!}{(n-k)!}$	3 of 8 runners: 336
k choices from n , repetition allowed	n^k	4-digit PIN: 10^4
arrange n with repeats $n_1, n_2, \dots$	$\frac{n!}{n_1! n_2! \dots}$	MATEMATIKA

$0! = 1$ IS A DEFINITION, AND A NECESSARY ONE

It makes $A(n, n) = \frac{n!}{0!} = n!$ come out right. There is exactly one way to arrange nothing.

Combinations

$$C(n, k) = \frac{n!}{k!(n-k)!}$$

Choosing k from n when **order does not matter**. It is $A(n, k)$ divided by $k!$, because each unordered choice was counted once for every ordering.

PROPERTY	STATEMENT
symmetry	$C(n, k) = C(n, n-k)$
ends	$C(n, 0) = C(n, n) = 1$
Pascal	$C(n, k) = C(n-1, k-1) + C(n-1, k)$
total	$\sum C(n, k) = 2^n$

Symmetry is a labour-saving device: $C(20, 18) = C(20, 2) = 190$, computed in one line instead of ten.

Does order matter?

THE ONE QUESTION TO ASK

Would swapping two of the chosen things give a **different** outcome? If yes, order matters — use $A(n,k)$. If no, use $C(n,k)$.

SITUATION	ORDER?	FORMULA
gold, silver, bronze from 8	yes	$A(8,3) = 336$
a team of 3 from 8	no	$C(8,3) = 56$
a 4-digit code	yes, repeats allowed	10^4
a hand of 5 cards from 36	no	$C(36,5)$
seating 6 people in a row	yes	$6! = 720$

02 Worked examples

EXAMPLE 1 From 8 runners, in how many ways can the first three places be filled?

1 Order matters — gold is not silver.

2 $8 \times 7 \times 6$

3 $= 336$

ANSWER

336

EXAMPLE 2 How many teams of 3 can be chosen from 8 people?

① Order does not matter.

②
$$C(8,3) = \frac{8 \times 7 \times 6}{3 \times 2 \times 1}$$

③
$$= 56$$

ANSWER

56

EXAMPLE 3 How many arrangements has the word MATEMATIKA?

① 10 letters: A×3, M×2, T×2, E, I, K.

②
$$\frac{10!}{3! \cdot 2! \cdot 2!}$$

③
$$= \frac{3\,628\,800}{24} = 151\,200$$

ANSWER

151 200

03 Interactive model

Ask “does swapping two of them change the answer?” before every formula.

Order or not?

Gold, silver and bronze from 8:

$$C(8,3)$$

$$A(8,3)$$

$$8^3$$

$$3!$$

A team of 3 from 8:

$$C(8,3)$$

$$A(8,3)$$

$$8^3$$

$$3!$$

$C(8,3)$ equals:

$$24$$

$$56$$

$$336$$

$$112$$

$C(20,18)$ equals:

A 4-digit PIN with repeats:

$0!$ equals:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Factorial	Faktorial	Факториал
Permutation	O'rin almashtirish	Перестановка
Arrangement	Joylashtirish	Размещение
Combination	Kombinatsiya	Сочетание
With repetition	Takrorlanish bilan	С повторениями
Order matters	Tartib muhim	Порядок важен

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

The multiplication rule is used with:

$C(n, k)$ equals:

\frac{n!}{(n-k)!}"/>

\frac{n!}{k!(n-k)!}"/>

n^k"/>

n!"/>

$C(n, k) = C(n, n-k)$ because:

Seating 6 in a row:

$\Sigma C(n,k)$ over all k is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $5!$

ANSWER 120

2. $0!$

ANSWER 1

3. $C(5,2)$

ANSWER 10

4. $C(6,3)$

ANSWER 20

5. $A(5,2)$

ANSWER 20

6. Arrangements of 4 books

ANSWER 24

7. 3-digit codes from 10 digits, repeats allowed

ANSWER 1000

Medium · the standard question

7 PROBLEMS

1. First three of 8 runners

ANSWER 336

2. Teams of 3 from 8

ANSWER 56

3. $C(20,18)$

ANSWER 190

4. Arrangements of MATEMATIKA

ANSWER 151 200

5. A menu: 4 starters, 6 mains, 3 desserts — full meals

ANSWER 72

6. Committees of 4 from 10

ANSWER 210

7. Arrangements of 6 people with 2 fixed together

ANSWER 240

Hard · stretch and reason

7 PROBLEMS

1. Hands of 5 from 36 cards

ANSWER $C(36,5) = 376\,992$

2. Arrangements of the 8 letters of ASSALOMU with the two S apart

ANSWER $\frac{8!}{2!} - 7! = 15\,120$

3. From 6 men and 4 women, committees of 4 with at least 1 woman

ANSWER 195

4. How many 4-digit numbers have all digits different?

ANSWER 4536

5. How many diagonals has a convex n -gon?

ANSWER $\frac{n(n-3)}{2}$

6. Prove $C(n,k) = C(n-1,k-1) + C(n-1,k)$

ANSWER Split on whether a fixed element is chosen

7. Prove $\sum C(n,k) = 2^n$

ANSWER Each element is in or out

07 Homework

Homework — 6 tasks

Write “order matters” or “order does not matter” before every formula.

1. Evaluate $7!$, $C(9,4)$, $A(9,4)$ and $C(15,13)$.
2. From 10 players, how many ways to pick a team of 5? And to pick a captain, vice-captain and three others?
3. How many arrangements has the word TASHKENT? And PARALLEL?
4. How many 5-digit numbers have all digits different and do not start with 0?

5. From 7 men and 5 women, how many committees of 4 contain at least 2 women?
6. Explain in three sentences how to decide whether order matters.